

MOCK INTERVIEW ASSESSMENT REPORT

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The interview followed the uploaded assessment instructions: exactly 10 questions covering Application Layer Protocols, Modern Communication Systems, Network Security, and scenario-based analysis.

INTERVIEW QUESTIONS AND RESPONSES

Q1

Question Asked: What is DNS (Domain Name System), and why is it important in computer networks? Explain how DNS helps a user access a website using a domain name instead of an IP address.

Student Response:

DNS is like the phonebook of the Internet. It converts a website's domain name into its IP address. DNS helps users remember easy domain names, helps computers find the correct web server, and makes accessing websites easier. For example, when www.google.com is entered, DNS finds its IP address and connects the computer to Google's server.

Q2

Question Asked: Compare SMTP and POP protocols. What is the role of each protocol in an electronic mail system, and when would you use them?

Student Response:

SMTP stands for Simple Mail Transfer Protocol and is used to send emails. POP stands for Post Office Protocol and is used to receive/download emails. SMTP transfers mail from the sender to the mail server and between mail servers, while POP downloads emails from the mail server to the user's device.

Q3

Question Asked: What is P2P communication? Explain how it works and give one real-world example.

Student Response:

P2P is a network where computers communicate directly with each other without depending on a central server. Each computer can send and receive data from other peers. BitTorrent is an example where users share files directly with other users.

Q4

Question Asked: What is VoIP? Explain how VoIP works and mention one practical application.

Student Response:

VoIP is a technology that allows voice calls over the Internet instead of traditional telephone networks. It converts voice into digital data packets and sends them through the Internet to the receiver. WhatsApp calls are an example of VoIP.

Q5

Question Asked: What is SSL? Explain how SSL helps protect data during communication over a network.

Student Response:

SSL is a security protocol that protects data sent between a user and a website. It encrypts the data so unauthorized people cannot easily read or change it during transmission. Online banking websites use SSL to protect login and account information.

Q6

Question Asked: What is a firewall? Explain how it protects a network from unauthorized access. Give one practical example.

Student Response:

A firewall is a security system that controls incoming and outgoing network traffic. It blocks unauthorized or harmful connections and allows only trusted traffic. A company can use a firewall to prevent unauthorized users from accessing its internal network.

Q7

Question Asked: A company notices that its employees can access websites normally, but the network becomes very slow when many employees use it at the same time. What could be causing this problem, and what network performance measures would you use to analyze it?

Student Response:

The network may be experiencing high traffic or bandwidth congestion because many employees are using it at the same time. The measures include bandwidth, latency, packet loss, and throughput.

Q8

Question Asked: A user can access a website, but the website is not loading securely and the browser shows a security warning. Which network security mechanism or protocol could be involved, and how would you troubleshoot the problem?

Student Response:

SSL/TLS (HTTPS) provides secure communication between the browser and website. Troubleshooting includes checking whether the website uses HTTPS, checking whether the SSL/TLS certificate is valid or expired, checking the device's date and time, and trying another browser or network.

Q9

Question Asked: A company wants to allow employees to communicate using voice calls over the Internet while maintaining good call quality and security. Which technology or protocols would you recommend, and why? Explain factors that can affect VoIP performance.

Student Response:

Use VoIP with SIP and SRTP. SIP establishes and manages voice calls, while SRTP encrypts voice data. QoS can give voice traffic higher priority. Factors affecting performance include bandwidth, latency, jitter, packet loss, and network congestion.

Q10

Question Asked: A company experiences a sudden increase in network traffic, and legitimate users are unable to access its services. How would you identify whether this could be a DoS attack, and what security mechanisms could be used to protect the network?

Student Response:

Check for a sudden unusual traffic increase, many requests from unknown sources, and services becoming unavailable to legitimate users.

QUESTION-WISE EVALUATION TABLE

Question	Technical Knowledge	Problem Solving	Analytical Thinking	Communication	Confidence
Q1	Level 4	Level 4	Level 4	Level 4	Level 3
Q2	Level 4	Level 4	Level 4	Level 4	Level 3
Q3	Level 4	Level 4	Level 4	Level 4	Level 3
Q4	Level 4	Level 4	Level 4	Level 4	Level 3
Q5	Level 3	Level 3	Level 3	Level 4	Level 3
Q6	Level 4	Level 4	Level 4	Level 4	Level 3
Q7	Level 4	Level 4	Level 4	Level 4	Level 3
Q8	Level 4	Level 4	Level 4	Level 4	Level 3
Q9	Level 4	Level 4	Level 4	Level 4	Level 3
Q10	Level 3	Level 2	Level 3	Level 3	Level 3

The rubric uses Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, and Level 1 = 25% of each criterion's weightage.

FINAL RUBRIC EVALUATION

Criterion	Weightage	Assigned Level	Marks Awarded
Technical Knowledge	25	Level 4	25
Problem Solving	25	Level 3	18.75
Analytical Thinking	20	Level 4	20
Communication	15	Level 4	15
Confidence	15	Level 3	11.25
Total	100		90.00

SUMMARY TABLE

Technical Knowledge (25)	Problem Solving (25)	Analytical Thinking (20)	Communication (15)	Confidence (15)	Sub
25	18.75	20	15	11.25	

OVERALL FEEDBACK

Strengths

- Demonstrated strong understanding of **DNS, SMTP, POP, P2P, VoIP, firewalls, SSL/TLS, and network performance metrics.**
- Gave appropriate real-world examples such as **Google DNS, BitTorrent, WhatsApp calls, online banking, and organizational firewalls.**
- Correctly identified important VoIP performance factors such as **latency, jitter, packet loss, bandwidth, and congestion.**
- Showed good ability to connect theoretical concepts with practical networking scenarios.

Areas for Improvement

- Q10 should include specific protection mechanisms such as **firewalls, traffic filtering/rate limiting, intrusion detection/prevention, and DDoS protection.**
- Expand answers slightly for analytical questions by explaining **why** a particular solution is appropriate.
- For security topics, use current terminology such as **TLS/HTTPS** rather than referring only to SSL.
- Continue developing confidence and giving structured, complete answers.

Interview Readiness

GOOD

Overall, the student demonstrated good technical knowledge and practical understanding of Computer Networks. The main improvement area is providing more complete security mechanisms and deeper reasoning in advanced scenario-based questions.